

Proteome-derived, database-searchable peptide libraries for identifying protease cleavage sites

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We introduce human proteome-derived, database-searchable peptide libraries for characterizing sequence-specific protein interactions. To identify endoprotease cleavage sites, we used peptides in such libraries with protected primary amines to simultaneously determine sequence preferences on the N-terminal (nonprime P) and C-terminal (prime P') sides of the scissile bond. Prime-side cleavage products were tagged with biotin, isolated and identified by tandem mass spectrometry, and the corresponding nonprime-side sequences were derived from human proteome databases using bioinformatics. Identification of hundreds to over 1,000 individual cleaved peptides allows the consensus protease cleavage site and subsite cooperativity to be readily determined from P6 to P6'. For the highly specific GluC protease, >95% of the 558 cleavage sites identified displayed the canonical selectivity. For the broad-specificity matrix metalloproteinase 2, >1,200 peptidic cleavage sites were identified. Profiling of HIV protease 1, caspase 3, caspase 7, cathepsins K and G, elastase and thrombin showed that this approach is broadly applicable to all mechanistic classes of endoproteases.

Peptide libraries are useful tools for determining the sequence specificity of ligand or substrate binding sites and enzymatic activities, including protease cleavage sites^{1,2}. Proteases sculpt the composition and functionality of all proteomes through specific processing or nonspecific degradation. With >559 members in humans, representing 5–10% of all drug targets³, the size of this enzyme family reflects the biological importance of proteolysis in post-translational protein modification⁴. MEROPS⁵, the peptidase database (<http://merops.sanger.ac.uk>), lists ~2,000 peptidases but only 7,000 cleavage sites. Indeed, numerous orphan proteases are annotated solely on the basis of sequence homology and have no functional characterization⁴. The cleavage site selectivity of peptidases involves recognition of amino acid residues spanning the scissile peptide bond. The Schechter and Berger nomenclature⁶ designates residues C-terminal to the cleavage site as prime-side (P') and N-terminal residues as nonprime-side (P). P and P' residues dock into the active site cleft in topographical and chemical complementary subsites designated S and S', respectively. Determining protease cleavage site consensus motifs is crucial for the design of sensitive and selective synthetic peptide-substrate cleavage assays. Such assays and the comprehensive mapping of active site specificity determinants are indispensable for developing protease inhibitor drugs^{7,8}.

Several library-based approaches are used to determine protease cleavage-site specificity, but all suffer from limitations. Although substrate phage display and other genetic screens^{9,10} identify cleavable sequences after several rounds of colony selection, they do not reveal the precise cleavage sites, which must be subsequently identified using a peptide designed from these sequences^{9–12}. Such multiple steps,

together with frequent sequencing of identical clones, are time consuming and decrease throughput. Positional scanning combinatorial peptide libraries systematically map active site selectivity, but only one position at a time^{13,14}. Their context-independent nature also precludes subsite cooperativity analysis^{1,2}, which can be studied using immobilized or spatially separated arrays of individual peptidic substrates as well as by nucleic acid-encoded peptide libraries^{15–18}. Here, proteolysis is typically detected by the release of N- or C-terminal fluorophores, thereby restricting these screens to either prime or nonprime sequences. This obstacle was overcome by first establishing the prime-side specificity of cleaved peptide pools¹⁹. A second peptide library is then designed, with all members anchored by the preferred prime-side residues. This two-step iterative workflow determines both prime and nonprime-side specificity. However, it is time consuming and the use of Edman sequencing of pooled libraries and not of individual peptides decreases the confidence of predictions with increasing cycles of sequencing and subsite cooperativity cannot be determined.

Our proteomic technique for determining protease cleavage sites relies on database-searchable oligopeptide libraries consisting of natural biological sequences derived from human proteomes in protease cleavage assays. By analysis of hundreds of cleaved sequences in a single experiment, the character of the active site subsite preferences and protease consensus cleavage site can be rapidly determined. The proteomic identification of protease cleavage sites (PICS) method selectively isolates carboxy-peptide cleavage products and uses liquid chromatography–tandem mass spectrometry (LC-MS/MS) to identify the prime side sequences of these cleaved peptides.

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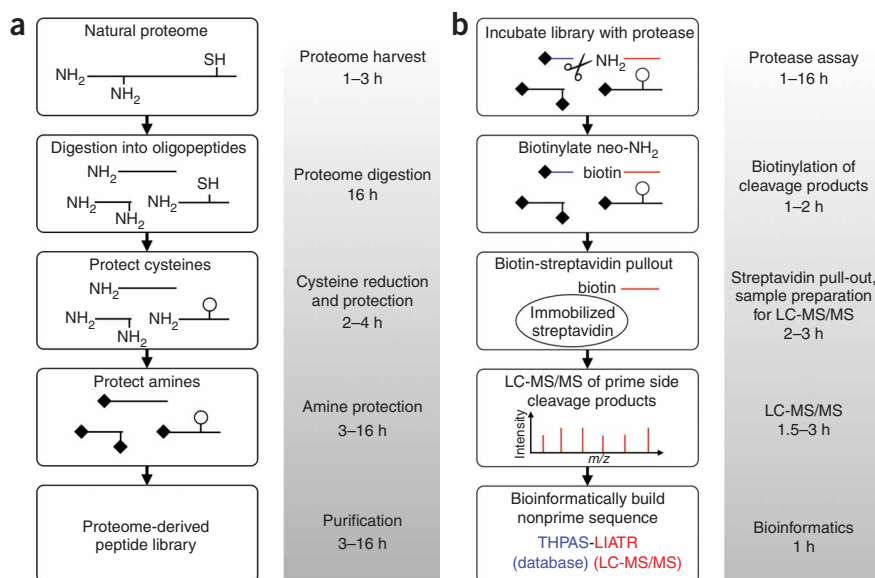


Figure 1 PICS peptide library generation and cleavage site screen. **(a)** Library generation. Proteome-derived peptide libraries are generated by digesting the proteome with trypsin, GluC or chymotrypsin to convert all proteins into oligopeptides. After inactivation of the digestion protease, cysteine sulfhydryls are protected with iodoacetamide to prevent cross-linking and chemical cross-reactivity. Primary amines are converted into tertiary amines by reductive methylation. **(b)** Cleavage site screen. After purification, the blocked peptide library is incubated with a protease of interest. Prime-side cleavage products have neo-amino termini that are conjugated to biotin through a cleavable disulfide bond. After isolation by immobilized streptavidin followed by mild (reducing) elution conditions, the sequences of the prime-side cleavage products are determined by LC-MS/MS (shown in red). The prime-side sequence is used to bioinformatically reconstruct the nonprime sequence of the cleaved peptide (shown in blue for the peptide THPASLIATR cleaved at SL) up to the first occurrence of a cleavage site of the protease used for library generation. The durations (h) provide an approximate timeline for a typical PICS experiment.

Nonprime-side sequences are automatically filled in by automated database searches of the human proteome. Hence, PICS simultaneously determines both prime and nonprime-side amino acid preferences from a single experiment. We validated PICS using GluC and caspases 3 and 7, which all possess canonical cleavage site motifs. Profiling of matrix metalloproteinase (MMP)-2, a protease with broad active site specificity, and a panel of serine, aspartic and cysteine proteases demonstrates the broad applicability of the approach.

RESULTS

Workflow

The PICS workflow is summarized in **Figure 1**. To generate peptide libraries, proteome samples are digested with trypsin (R or K in P1), GluC (E or D in P1) or chymotrypsin (W or Y or L or F in P1). After protease inactivation, peptide sulfhydryl groups are carboxyamido-methylated and amino termini and lysine ϵ -amino groups are dimethylated. The protease of interest is incubated with these amine-protected peptide libraries of natural biological sequences under appropriate conditions (pH, buffer, temperature). Proteolysis generates unprotected neo-amino termini on carboxy prime-side cleavage products. These are tagged with cleavable biotin using an *N*-hydroxysuccinimide ester as an amine-reactive group, then isolated with immobilized streptavidin, and identified by LC-MS/MS to yield the prime-side sequence of the cleavage site. As proteome-derived peptide libraries are database searchable, the corresponding nonprime-side sequences (N-terminal to the scissile bond) are obtained using bioinformatics based upon the prime-side sequence and recorded until reaching the next cleavage site of the protease used for peptide library generation.

Although nonprime sequences can be assigned with high confidence for most cleaved peptides, bioinformatics yields ambiguous sequences for certain peptides. For example, the MMP-2 prime-side cleavage product VSVVDLTCR occurs in eight proteins (see **Supplementary Table 1** online). For some proteins, P1 is aspartate whereas in others P1 is asparagine, so only the P4–P2 sequence is used to determine subsite preferences. This procedure is automated with Perl computer scripts (**Supplementary Scripts and Note** online). For a peptide with an N terminus that matches the specificity of the enzyme used to prepare the library, it is impossible to distinguish whether this peptide originates from incomplete amine protection or cleavage by the test protease at a site missed by the enzyme used to make the library. Hence, such peptide identifications are excluded from further analysis and libraries prepared with proteases of different specificity should also be used.

The large cohort of individual cleavage site sequences is used to determine subsite specificity and cooperativity by multiple sequence alignments and are displayed as heat maps or, after correction for natural amino acid occurrence rates, as sequence logos^{20,21}. PICS peptide libraries have a very similar amino acid composition as the human proteome in the International Protein Index (IPI)²² (**Supplementary Fig. 1** and **Supplementary Table 2**

online). Hence, in determining the protease cleavage site consensus motifs, the IPI amino acid frequencies are used to correct for natural amino acid occurrence rates. Using PICS, hundreds to thousands of different cleavage site sequences, often from P6–P6', can be simply and rapidly identified and the protease subsites and cleavage site consensus motifs mapped.

Validation

We validated PICS using GluC (*Staphylococcus aureus* protease V8), which specifically cleaves with P1 glutamate specificity and to a lesser extent, aspartate²³. Of the GluC-cleaved peptides from a tryptic library, 94–96% matched the expected GluC specificity (**Figs. 2a–c** summarize the peptide sequences listed in **Supplementary Tables 3** and **4** online). As the GluC specificity resides in the nonprime position P1, the high rate of correct GluC cleavage site sequences validates the bioinformatic derivation of the nonprime-side sequences. Indeed, the P1 identity remained ambiguous in only 1% of all cases, all of which were cleavage sites where several different nonprime-side sequences could be assigned to the prime-side product. Only 10–15 peptides with tryptic or mature protein amino termini were identified. Identification of <10 peptides in the absence of a test protease in routine control experiments confirms essentially full amine protection.

Counterintuitively, in comparing GluC/library ratios, fewer cleaved peptides (297) were identified at an enzyme/library ratio of 1:100 than at 1:1,000 (453 cleaved peptides). The higher GluC concentration yielded a more complete digest resulting in more prime-side peptides that were too short for MS/MS identification. Indeed, prime-side sequences contained fewer glutamate residues, indicating fewer missed

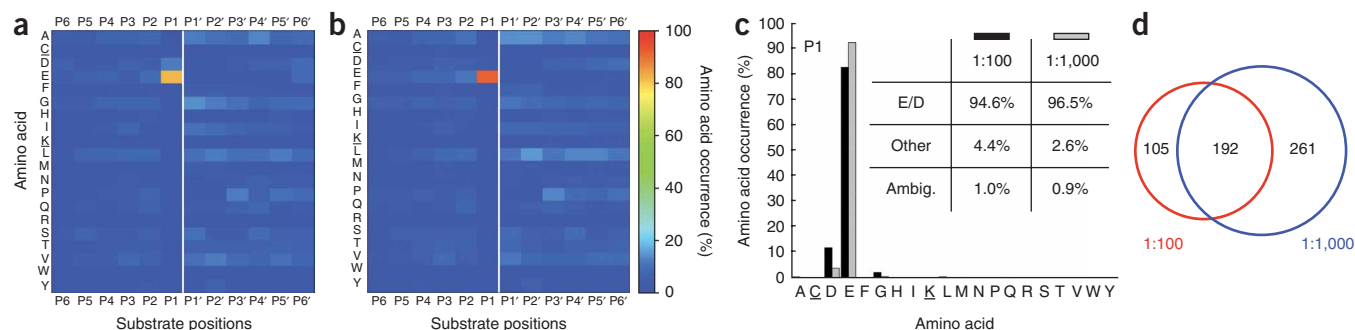


Figure 2 GluC cleavage sites of a tryptic PICS library. A tryptic PICS library was generated from the lysate of cultured HEK cells and incubated with GluC at ratios of 1:100 and 1:1,000 (enzyme to library, 200 μ g of PICS library). At both ratios, >94% of the identified cleavage sites match the expected GluC specificity. Other than E or D in P1, no selectivity for any other residue between P6–P6' was observed. We identified over 290 unique cleavage sites for the GluC/library ratio of 1:100 and over 450 unique cleavage sites for the GluC/library ratio of 1:1,000 (cleavage sites are listed in **Supplementary Table 3** and **4**). (a,b) Amino acid occurrences for P6–P6' in the form of a heat map. GluC/library ratio of 1:1,000 (a). GluC/library ratio of 1:100 (b). (c) Amino acid occurrences in P1. (d) Distribution of unique and shared cleaved tryptic peptides found at both enzyme/library ratios. *ambig.*, signifies ambiguous; that is the prime-side sequence could be assigned to different tryptic peptides with different sequences, resulting in ambiguous identifications of the P1 residue. C, carboxyamidomethylated cysteine; K, dimethylated lysine. The scissile peptide bond between P1 and P1' is indicated by a white line. Heat maps were created using pro Fit (QuantumSoft). The area proportional Venn diagram was drawn using Venn Diagram Plotter from the Pacific Northwest National Laboratory (<http://ncrr.pnl.gov/software/VennDiagramPlotter.stm>).

GluC cleavage sites (**Supplementary Fig. 2** online). Nonetheless, 65% of peptides identified at an enzyme/library ratio of 1:100 were also identified at the ratio of 1:1,000 (**Fig. 2d**).

Complementary peptide libraries increase coverage

MMP-2, an important protease in tumor metastasis and angiogenesis³, was used as a prototypical protease that lacks a canonical cleavage site. Analysis of its active site-specificity by phage display¹¹, cleavage of synthetic peptides^{24,25} and mixture-based oriented peptide libraries¹⁹ provides a benchmark to further validate PICS. MMP-2 has loose active site specificity, with each subsite accommodating several amino acids. Leucine in P1' is the major specificity determinant but subsite preferences are also found, notably proline at P3 (ref. 19).

To reduce any peptide sequence bias, we used four different tryptic peptide libraries to map the active site selectivity of MMP-2. Separate

libraries were generated from human embryonic kidney (HEK) cell lysates, human HT1080 fibrosarcoma cell lysates and detergent-solubilized HEK cell lysis pellets. In total, 1,243 cleaved peptides were identified, with 1,179 possessing different P3–P3' sequences (**Supplementary Tables 1** and **5–7** online). Sixty-four cleavage-site sequences (P3–P3') occurred more than once but in different tryptic peptides. Notably, leucine was present in 46% of all P1' positions (**Fig. 3a**). Whereas the four individual PICS experiments yielded comparable MMP-2 specificity profiles (**Supplementary Fig. 3** online), 66% of the cleaved peptides were found in only one tryptic library and 34% were identified in two or more tryptic libraries (**Supplementary Fig. 4** online). Hence, although PICS yielded consistent specificity profiles from peptide libraries with different sequence composition, the use of multiple libraries in parallel substantially increased cleavage site coverage and reflects the

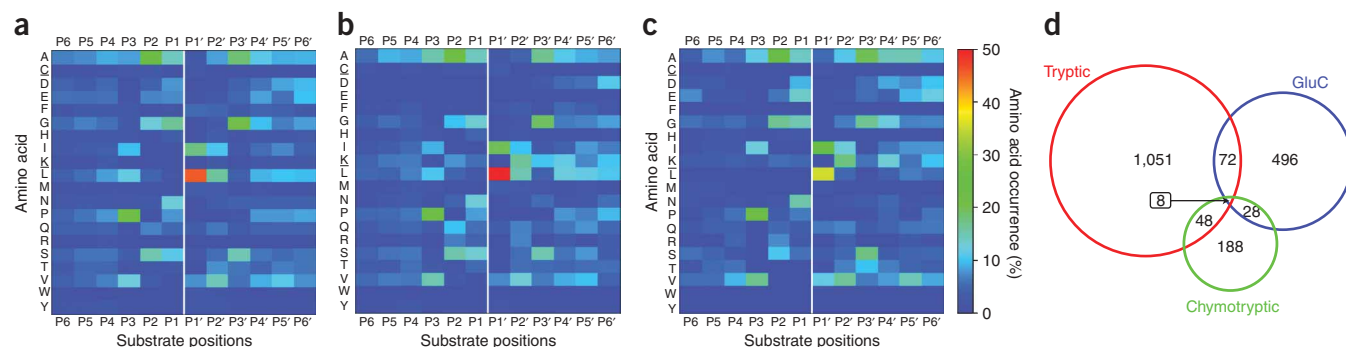


Figure 3 MMP-2 cleavage sites in peptide libraries. (a) Tryptic peptide libraries. (b) GluC peptide libraries. (c) Chymotryptic peptide libraries. PICS libraries were incubated with MMP-2 at a ratio of 1:50 (enzyme/library). Four different tryptic, four different GluC and four different chymotryptic PICS libraries were used (each 200 μ g). For each class of PICS library, the average amino acid occurrences in P6–P6' were calculated from the four experiments and are shown in the form of a two-dimensional heat map. Both the tryptic and GluC libraries confirm the predominant MMP-2 specificity for leucine in P1' and display a high level of consistency. Other specificities are located between P3–P3', whereas no selectivity is observed outside P3–P3', other than a weak selectivity at P4'. C, carboxyamidomethylated cysteine; K, dimethylated lysine. The scissile peptide bond between P1 and P1' is indicated by a white line. (d) Whereas the different library classes show very similar specificity profiles, the actual MMP-2 cleavage site sequences in P3–P3' from the different library classes show little overlap. Of the 1,243 identified tryptic peptides that were cleaved by MMP-2, 1,179 had different P3–P3' sequences. Similarly, of the 633 GluC peptides cleaved by MMP-2, 604 possess different P3–P3' sequences and the 286 chymotryptic peptides cleaved by MMP-2 comprise 272 different P3–P3' sequences.

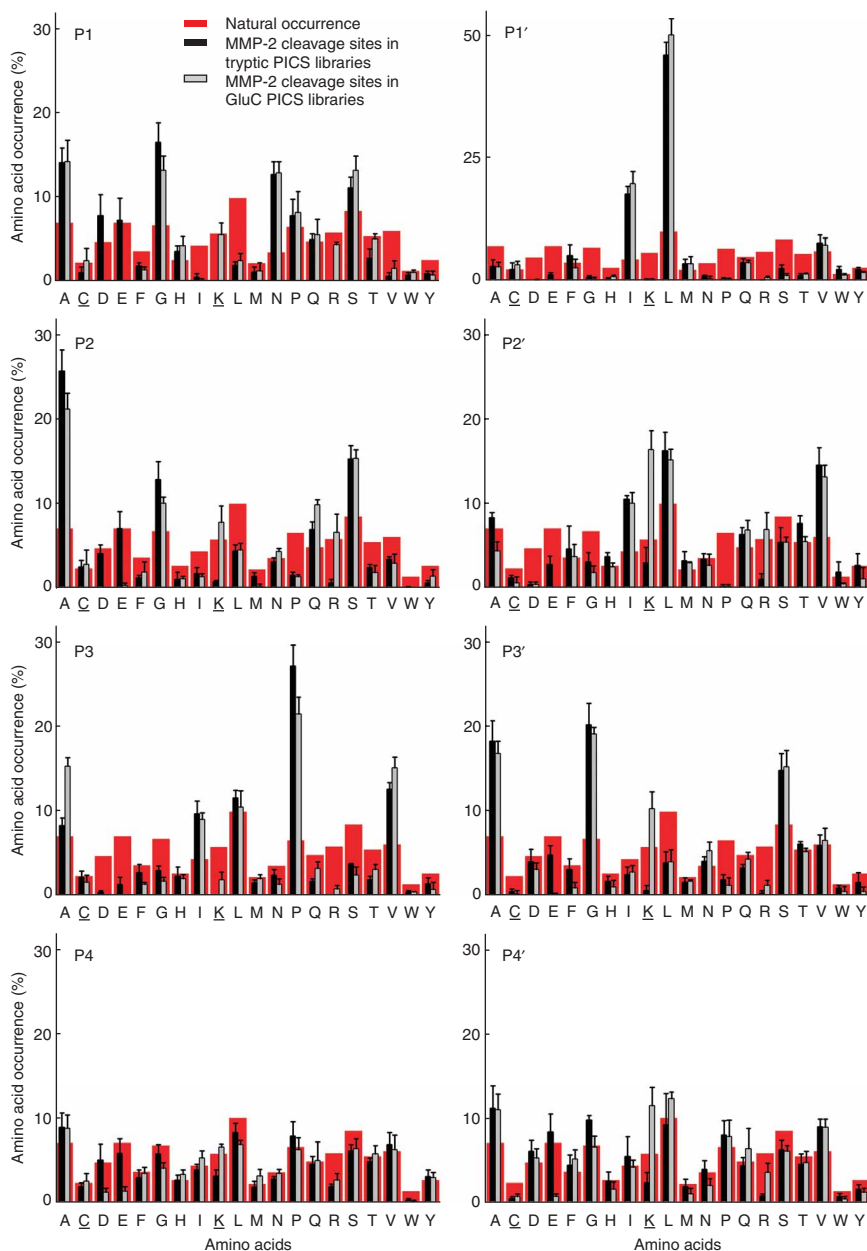


Figure 4 Amino-acid occurrences in P4–P4' for MMP-2 cleavage sites in tryptic and GluC peptide libraries. MMP-2 cleavage sites were identified in tryptic or GluC peptide libraries as described for **Figure 3**. Shown is the average amino acid occurrence in P4–P4' found using the four tryptic libraries (black bars) and the four GluC libraries (gray bars). (Error bars, s.d.). The average amino acid occurrences are shown in relation to the natural amino acid composition (red bars, see also **Supplementary Fig. 1**). Whereas the amino acid ratios in P4' approach the natural occurrence, dimethylated lysine and alanine are present at slightly elevated levels indicating a previously unobserved minor preference. The amino acid occurrences of the individual experiments are presented in **Supplementary Figures 3 and 5**; cleavage site sequences are listed in **Supplementary Tables 1 and 5–11**. 200 µg of each PICS library were used at an enzyme/library ratio of 1:50.

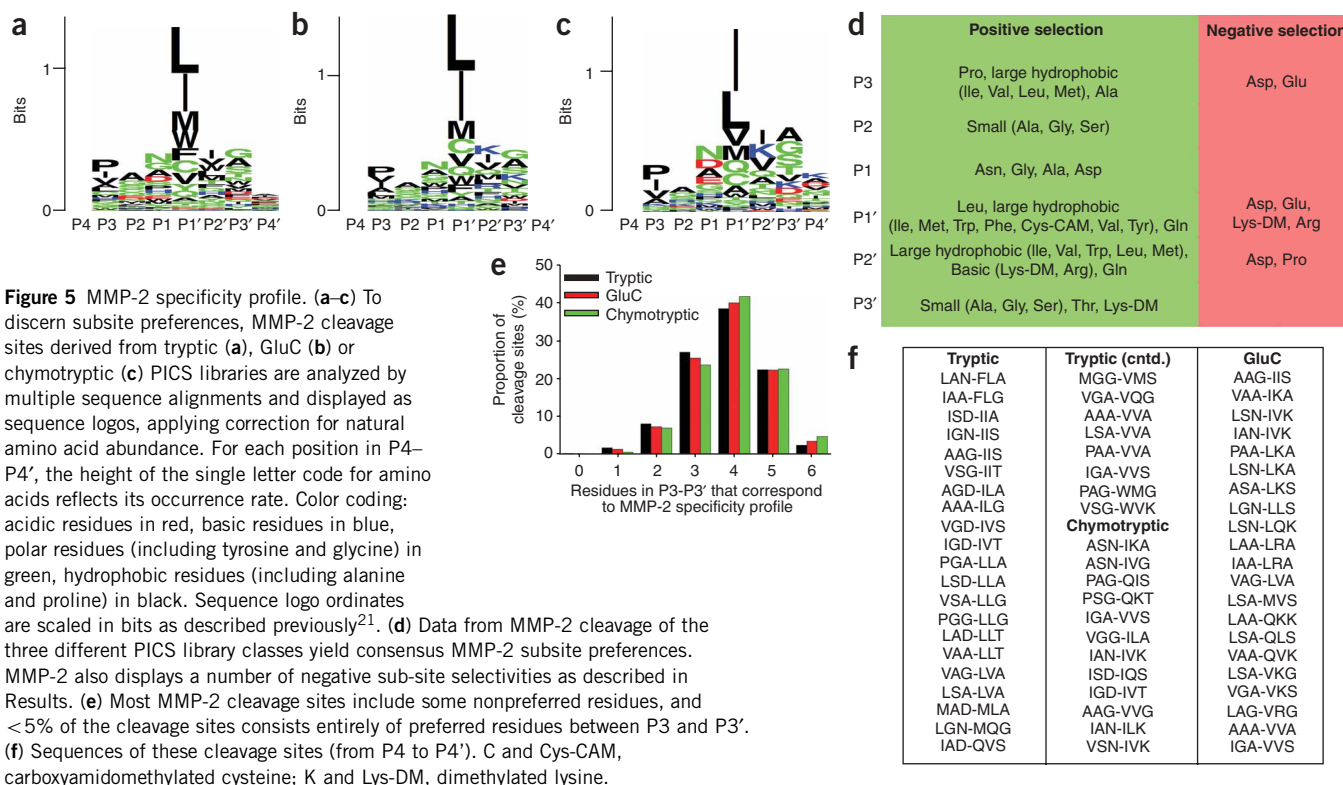
(**Figs. 3 and 4**) although only 80 identical cleavage sites between P3–P3' were found in both library classes. This shows that a large number of individual sequences can be analyzed with this approach (**Fig. 3d**).

To illustrate the limitations of using a library generated from digestion of a proteome with a protease that has overlapping specificity to the test protease, we used chymotryptic peptide libraries. Chymotrypsin has a P1 specificity for large hydrophobic residues (leucine, phenylalanine, tyrosine and tryptophan), whereas MMP-2 prefers large hydrophobic residues in P1' (**Fig. 3**). This conflicting specificity reduced the number of identified cleaved peptides to 286 in four separate PICS experiments involving different chymotryptic libraries. Even so, chymotrypsin-generated peptide libraries corroborate the positional occurrences observed for the tryptic and GluC peptide libraries. In particular, 35% of cleavage sites had leucine in P1, which were detected owing to missed chymotryptic cleavages at leucine during library generation (**Fig. 3c**; sequences in **Supplementary Tables 12–15** online).

Proteome digestion with trypsin, GluC and chymotrypsin yields different peptide library sequences, different average peptide lengths and thereby minimizes the possibility that incomplete digestion of any protein causes a particular sequence to be underrepresented. In any case, peptides with missed cleavage sites remain available for cleavage within the library. Although tryptic peptides lack internal basic residues and GluC peptides lack internal glutamate residues, chymotryptic peptides have both internal acidic and basic residues but a lower content of internal hydrophobic residues. Therefore, protease profiling using three library classes minimizes systematic bias due to amino acid positioning relative to the peptide termini, secondary structure and differing peptide lengths. Hence, proteome-derived peptide libraries are sufficiently rich in sequence diversity to reduce such bias.

undersampling difficulties in obtaining complete coverage in proteomic experiments.

As tryptic peptides lack internal lysine or arginine residues and so cannot be used to probe subsite preferences for basic amino acids, peptide libraries were generated using GluC, which cuts C-terminal to glutamate and aspartate²³ (**Fig. 2**). Four separate GluC peptide libraries were prepared from cell lysates and detergent-solubilized lysis pellets of HEK cells. MMP-2 cleavage of these libraries yielded 633 unique cleaved peptides, with 604 having different P3–P3' sequences. Forty-eight percent of cleavage sites had a P1' leucine (**Fig. 3b**; individual sequences in **Supplementary Tables 8–11** online). As observed for the tryptic peptide libraries, MMP-2 cleavage of four different GluC libraries yielded similar specificity profiles (**Supplementary Fig. 5** online), although most cleavage sites were identified only once (**Supplementary Fig. 4**). Notably, the MMP-2 specificity profiles from tryptic and GluC libraries were highly convergent



MMP-2 cleavage site specificity

Although the P3–P3' cleavage site sequences from the three different library classes showed little overlap (Fig. 3d), consistent specificity profiles were observed independent of the proteome source or the digestion enzyme used (Fig. 4 for tryptic and GluC libraries and Supplementary Fig. 6 online for chymotryptic libraries). Outside of P3–P3', the amino acid ratios approach the natural amino acid composition, thus confirming the nonrandom occurrence of elevated amino acid levels in P3–P3'. This analysis also reveals the spatial limits of the active site in very good agreement with structural characterization of MMPs that show a substrate binding cleft extending from P3–P3' (refs. 26,27). However, we noted a previously undescribed weak substrate interaction at P4', as discussed below (Fig. 4 and Supplementary Fig. 6).

To map subsite preferences, we aligned the cohort of MMP-2 cleavage sites according to the prime and nonprime positions and analyzed these as sequence logos (Fig. 5a–c), taking into account the natural amino acid abundance. The PICS-based MMP-2 specificity profile (Fig. 5d) is in excellent agreement with studies that used mixture-based oriented peptide libraries¹⁹. In P3, PICS confirmed the well-characterized MMP-2 preference for proline and additionally showed MMP-2 selectivity for large hydrophobic residues. P2 had the lowest level of selectivity, but with a preference for small amino acids (alanine, glycine, serine) (Fig. 5a–c). The P1 preferences included the small amino acids alanine and glycine, but, interestingly, also asparagine and aspartate. Leucine in P1' (46%) forms the predominant molecular specificity of MMP-2 along with other large hydrophobic residues such as isoleucine, methionine, valine, phenylalanine or carboxyamidomethylated cysteine. In P2', the MMP-2 preferences included basic amino acids, notably dimethylated lysine formed during library preparation, and large hydrophobic residues. The P3' preferences included the small amino acids alanine, glycine

and serine. Although the amino acid ratios in P4' approached the natural occurrence (Fig. 4 for GluC and tryptic libraries, Supplementary Fig. 6 for chymotryptic libraries), substantially elevated levels of dimethylated lysine and alanine were observed. Negative selectivity was also observed: aspartate and glutamate were found at very low levels in P3, together with low levels of charged amino acids (aspartate, glutamate, arginine and dimethylated lysine) in P1' and of aspartate and proline in P2'.

Fewer than 5% of all MMP-2 cleavage sites had preferred residues in all six positions between P3–P3', with most cleavage sites possessing three to five preferred residues (Fig. 5e,f). This indicates that preferred residues in one position can compensate for the presence of residues with neutral or even negative preference in other positions. The close match of a simulated PICS experiment for MMP-2 cleavage at leucine in P1' on an *in silico* tryptic PICS library confirms that the PICS experimental specificity profile describes enzyme selectivity without influence from any repetitive sequence clusters in the library that might distort it (Supplementary Material and Supplementary Fig. 7 online). Overall, PICS demonstrates that MMP-2 selectivity is guided by subsite preferences rather than stringent consensus motifs such as for GluC or caspases^{23,28}.

Broad applicability of PICS

Profiling serine proteases using PICS identified 98 thrombin cleavage products from one GluC library, of which 91% have the known specificity for arginine in P1 (Fig. 6a and Supplementary Table 16 online). Proline was preferred in P2 (23%), and in P1' the preference was for serine (18%) and the two small amino acids alanine and glycine (combined total 38%). Although the 48 thrombin cleavage sites reported in MEROPS⁵ indicate a stronger selectivity for P1' serine than we found, the use of a considerably larger cohort of cleaved sequences generated in a single PICS experiment enabled greater

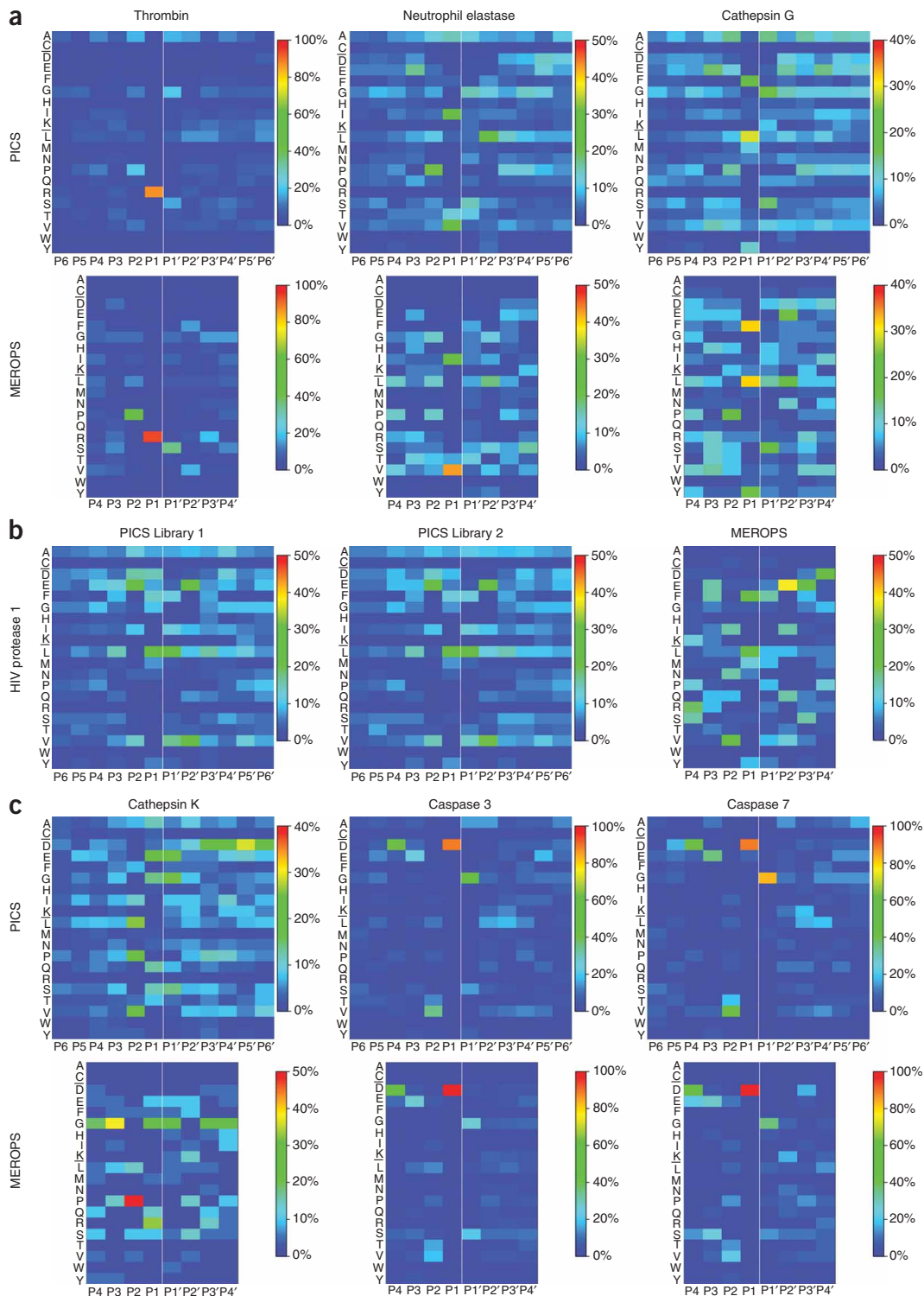


Figure 6 Application of PICS to serine, aspartic and cysteine proteases. **(a–c)** One PICS experiment for each of several serine **(a)**, aspartic **(b)** and cysteine **(c)** proteases was used to determine the general applicability of the method. For each protease, the PICS specificity profile is shown in a heat-map style together with a heat-map representation of the corresponding specificity profile reported in MEROPS⁵. For thrombin, PICS identified 98 cleavage products in a GluC library (sequences listed in **Supplementary Table 16**). For neutrophil elastase and cathepsin G, PICS identified 234 and 225 peptidic cleavage sites, respectively (sequences listed in **Supplementary Tables 17 and 18**). Specificity of the aspartic protease HIV protease 1 was profiled with two tryptic PICS libraries, yielding 304 and 463 cleavage products, respectively (sequences listed in **Supplementary Tables 19 and 20**). For the cysteine protease cathepsin K, PICS identified 132 cleavage sequences (listed in **Supplementary Table 21**). PICS profiling of caspases 3 and 7 yielded 55 and 37 cleavage products, respectively (sequences listed in **Supplementary Tables 22 and 23**). **C**, carboxamidomethylated cysteine; **K**, dimethylated lysine. All enzyme/library ratios were 1:100; 200 μ g library were used for thrombin, cathepsin G and elastase; 400 μ g library were used for HIV protease 1, cathepsin K, caspase 3 and caspase 7. Tryptic libraries were used for all proteases except thrombin, in which a GluC peptide library was used.

confidence in statistical analysis of subsite specificity. Indeed, the PICS P1' data agree with a biochemical analysis of thrombin specificity²⁹. A single PICS analysis identified 234 and 225 peptidic cleavage sites for neutrophil elastase and cathepsin G, respectively (Fig. 6a, Supplementary Tables 17 and 18 online). In good agreement with the 38 and 149 cleavage sites in MEROPS⁵, PICS distinguished elastase specificity for P1 isoleucine (27%) and valine (30%) from the cathepsin G specificity for P1 leucine (29%), phenylalanine (23%) and tyrosine (12%).

Aspartic HIV protease 1 specificity was profiled in two PICS experiments using different tryptic libraries, yielding 304 and 463 cleavage products (Fig. 6b, Supplementary Tables 19 and 20 online). Specificity profiles from both experiments agreed well with MEROPS⁵ (32 sites), with a partial preference for glutamine in P2' together with phenylalanine and leucine in P1. HIV protease 1 can cleave at P1' proline, leaving a secondary N-terminal amine, which is less reactive to N-hydroxysuccinimide esters than primary amines. Although this might limit identification of cleavage products with P1' proline, the two PICS screens identified 1.3% and 1.6% P1' proline, respectively. In comparison, PICS found no proline at this position for GluC, a protease that does not cleave at P1' proline (Fig. 2). In comparison to all HIV protease cleavage sites identified by PICS, peptidic substrates with P3 leucine possess higher levels of alanine or glycine in P1. Both PICS experiments yielded 15% P3 leucine and 8% P1 alanine. However, peptides containing a P3 leucine possess 22–25% P1 alanine alongside a decrease in P1 phenylalanine from 13–14% to 4–7%. Correspondingly, peptides containing a P1 alanine display higher levels of P3 leucine (39–46% compared to 15%), whereas peptides containing a P1 phenylalanine display a lower level of P3 leucine (5–7%) compared to 15% occurrence for all HIV 1 protease cleavage sites. Hence, by analyzing individual cleaved peptide sequences in large numbers, PICS readily reveals subsite cooperativity in sequence specificity.

Cysteine proteases were also profiled. PICS identified 132 cathepsin K cleavage sequences (Fig. 6c and Supplementary Table 21 online). Most of the 21 sites in MEROPS are derived from collagen cleavages³⁰ and therefore vastly overrepresent proline and glycine (at P2 and P3, respectively), which PICS showed to be more subtle preferences. However, both PICS and MEROPS⁵ show a preference for glycine in P1 and glutamate in P1'. Notably, PICS revealed a novel specificity requirement for aspartate in P5' outside the active site³⁰. This can be attributed to substrate interaction with a cationic patch at the 'back-side' of cathepsin K. For caspases 3 and 7, PICS identified 55 and 37 cleavage products, respectively, that accurately reflect the canonical nonprime-side selectivity, DEVD²⁸, together with a preference for glycine or serine at P1' and a previously unrecognized P2' preference for aliphatic hydrophobic residues, including alanine, leucine and valine (Fig. 6c, Supplementary Tables 22 and 23 online).

DISCUSSION

Proteome-derived peptide libraries from any fully sequenced species contain an enormous array of natural, biological sequences that can be applied to determine the sequence specificity of enzyme and protein/ligand interactions and post-translational modifications that are amenable to LC-MS/MS identification. We have demonstrated their use in the characterization of site-specific proteolysis in all mechanistic classes of human endoproteases with few limitations. Through a simple 1–2 d workflow (Fig. 1b) with <2 h of LC-MS/MS per sample, followed by rapid automated data analysis using the computer scripts provided (Supplementary Scripts and Note), hundreds of prime and nonprime sequences of individual cleaved peptides can be identified per assay.

PICS libraries can be prepared in batches, thus uncoupling library generation and protease profiling and thereby enabling the parallel screening of several proteases with multiple libraries. With improved access to MS facilities and automated data analysis, this should facilitate the use of proteome-derived peptide libraries.

We demonstrate >95% accuracy for PICS in resolving the canonical specificity of GluC and the extended nonprime-side selectivity of caspases 3 and 7 (ref. 28). Similarly, for a panel of metallo-, serine-, aspartic- and cysteine-proteases, PICS yielded specificity profiles in excellent agreement with existing data and further extended the specificity requirements to both prime and nonprime-sides. For example, oriented synthetic peptide libraries and PICS describe highly similar MMP-2 selectivity for P3–P3', with PICS alone finding a preference for asparagine in P1. This is corroborated by MMP-2 cleavage sites in human aggrecan (PEN-FFG)³¹ and the human pro-MMP-2 autolytic activation site (VAN-YNF)³². Hence, the tight correspondence of our data with other approaches^{11,19,24,25} further validates PICS. We also discovered interactions outside of the active site at P4' in MMP-2 and at P5' in cathepsin K. Similarly, the PICS profile predicts the span of the active site cleft by analysis of the subsites where the amino acid specificity occurrences return to the natural abundance levels. This is useful for enzyme characterization, particularly for proteases with no known three-dimensional structure, that might also assist in rational drug design.

A synthetic peptide of an MMP-2 cleavage site identified by PICS, and containing a lysine (underlined) in P2' (RGPANLKGR), was incubated with MMP-2. An otherwise identical synthetic peptide in which the lysine was dimethylated, as occurs during PICS library generation, was cleaved with equivalent kinetics (Supplementary Fig. 8 online). However, trypsin-like proteases, which do not cleave modified lysines, represent a limitation of the approach. Another synthetic peptide, RGPANCIGR, was not cleaved by MMP-2, but carbamido-methylation of the cysteine enabled MMP-2 cleavage between the modified cysteine (P1') and asparagine (P1). Although these amino acid modifications can alter specificity, in so doing they also reveal the importance of these interactions or sites in substrate recognition. We are developing proteome-derived peptide libraries employing unmodified cysteine and lysine residues (Supplementary Figs. 9 and 10 online). Few approaches can determine exopeptidase specificity due to bulky fluorophores or blocked N- or C-termini that interfere with substrate recognition. PICS libraries are not designed to profile exopeptidase activity as a dimethylated amino terminus usually prevents amino exopeptidase activity. In addition, carboxyl-cleaved single amino acids can not be directly detected using mass spectrometry by our approach.

Most cleavage-site screens focus on nonprime-side specificity but few approaches can determine prime-side selectivity^{1,2}. The latter requires multiple experiments, or can be performed only on a peptide-by-peptide basis³³, precluding a high-throughput analysis. In comparison, PICS simultaneously characterizes prime- and nonprime specificities of hundreds of peptides in the same experiment. The key to this advantage is the use of proteome-derived peptide libraries. Their database-searchable characteristic enables the automated reconstruction of the entire cleavage site sequence based on prime-side sequence information without further experimentation. In contrast to PICS, synthetic degenerate peptide libraries are not amenable to LC-MS/MS identification for two reasons: insufficient chemical differences between peptides prevents efficient LC separation and, critically, degenerate libraries also lack database-searchable sequences, which are required for efficient spectrum-to-sequence assignment³⁴.

PICS uses highly diverse biological sequence libraries that contain peptides of variable length which can improve peptide recognition, as shown for defining the extended P5' specificity site in cathepsin K. Although secondary structure formation and interactions distant to the active site cleft might potentially distort cleavage, the use of different proteome digestion enzymes or different proteome sources to prepare libraries overcomes such bias. Moreover, highly abundant cleavage products are analyzed only once by LC-MS/MS—identical peptides, no matter their abundance, co-elute from the sample column, then ionize and fly together in the mass spectrometer to give a single peptide ion peak. This is then subjected to MS/MS fragmentation only at this unique point in the mass spectrometer analysis. By contrast, in genetic screens, multiple cloning cycles and repetitive sequencing of the same clonal sequence limits coverage and is time consuming^{9,10}.

Substrate phage-display only provides sequences that encompass a cleavage site thus necessitating post-screening experiments to locate the precise scissile bond. In reality, this can generate only a restricted set of cleavage information. For MMP-2, substrate phage display identified <50 cleavable peptides¹¹, whereas a single PICS experiment for MMP-2 yielded >300 peptidic substrate sequences, including the precise location of the scissile peptide bond. This aspect is also an advantage over pooled synthetic peptide library approaches. By identifying hundreds to >1,000 individual cleavage sites, PICS enables robust statistical analysis of subsite preferences and subsite cooperativity. For instance, our demonstration for HIV protease 1 that a P1 alanine occurs at a threefold higher rate with a P3 leucine than found with other residues is of mechanistic interest that could be useful in drug development, as is the identification of individual cleaved peptides, which enables a seamless transition to design synthetic peptide cleavage assays.

PICS does not aim to identify physiological native substrates. Our approach is peptide-centric and differs substantially from degradomics approaches³⁵ used to characterize protease substrate degradomes^{33,36–39}. As these methods discover native natural substrates, they are protein-centric and are not usually well suited to perform subsite mapping. To prevent a distorted active site specificity profile, this is best defined using denatured peptide libraries that focus on the active site-mediated contribution. Structural features of intact proteins shield potential cleavage sites, whereas exosites may increase cleavage of less preferred sites⁴⁰. For example, the chemokine CCL7 is cleaved only at Gly⁴-Ile⁵ by MMP-2 (ref. 41), but denatured CCL7 is cleaved at multiple positions (data not shown). Hence, peptide-centric approaches expose more cleavable sequences to proteolysis than protein-centric ones, thereby generating larger quantities of specificity information. This is essential to robustly profile active-site specificity.

As for any peptide library screen, the cleavage sites identified by PICS do not necessarily correspond to their likelihood of cleavage in native proteins. Predictive bioinformatics utilities like PoPS (prediction of protease specificity) (<http://pops.csse.monash.edu.au/>) are a “reasoning environment for protease activity”⁴², which aim to predict protein substrates based on active-site specificity matrices. A PoPS analysis based on cleavage sites that correspond to the MMP-2 specificity profile in at least four of six positions predicts >5,000 MMP-2 cleavage sites in extracellular and membrane-associated human proteins. Although this includes a number of known MMP-2 substrates, such as fibronectin, collagens, decorin, osteonectin, procollagen C-proteinase enhancer, vascular endothelial growth factor D and connective tissue growth factor^{37,43}, such predictions suffer from an unacceptably high false-discovery rate. Hence, PICS speci-

city profiles, like any cleavage site data, should be used with great caution to screen for physiological substrates by database searches.

We are developing applications to use proteome-derived peptide libraries to pan for recognition motifs involved in receptor or cell-specific interactions, and as screens for sites of nonproteolytic, site-specific post-translational modifications such as phosphorylation. Using proteome-derived peptide libraries to combine natural proteomic sequence diversity with LC-MS/MS and database searches offers an approach to numerous screening applications.

METHODS

Cell culture. The human fibrosarcoma cell line HT1080 was grown in roller bottles in Dulbecco's modified Eagle's medium (Invitrogen) and 10% cosmic calf serum (HyClone). At 50% confluency, cells were washed extensively with PBS (138 mM NaCl, 2.7 mM KCl, 20 mM Na₂HPO₄, 1.5 mM KH₂PO₄, pH 7.4) to remove serum proteins and grown serum-free for 12 h. Cells were then washed three times and incubated with phenol red-free, serum-free medium for 48 h. Cells were then washed again and detached in PBS/0.2% (wt/vol) EDTA. The 293 HEK cell line was grown in suspension (5 liters) using the chemically defined, protein-free CD 293 medium (Invitrogen).

Peptide library preparation. Cells were centrifuged and resuspended in 100 mM 4-(2-hydroxyethyl)piperazine-1-ethanesulfonic acid (HEPES), pH 7.4, 150 mM NaCl. To prepare cell lysates, we disrupted the cells by nitrogen cavitation, followed by centrifugation at 40,000g, 1 h. After centrifugation, peptide libraries were generated from the soluble cytosolic fraction and the insoluble lysis pellet. The soluble fraction was heat-denatured at 70 °C, 20 min with addition of 0.02% SDS (wt/vol) and 5 mM dithiothreitol (DTT). After cooling, the soluble fraction was digested by trypsin (1-chloro-3-tosylamido-4-phenyl-2-butanone treated, Sigma-Aldrich), or GluC (Worthington), or chymotrypsin (1-chloro-3-tosylamido-7-amino-2-heptanone treated, Worthington) for 16, 37 °C at an enzyme/proteome ratio of 1:50.

The lysis pellet was resuspended in 100 mM HEPES pH 7.4, 0.02% SDS (wt/vol), 5 mM DTT, followed by denaturation at 80 °C, 30 min. After cooling, 10 µg/ml DNase I (Roche) was added and the suspension was incubated with either trypsin, GluC or chymotrypsin for 16 h, 37 °C. The protease-treated suspension was centrifuged (40,000g, 1 h) with the supernatant being used for peptide library preparation. In all preparations proteases were deactivated by 0.5 mM phenylmethylsulfonyl fluoride (30 min, 22 °C) followed by heat treatment (70 °C, 30 min) in the presence of additional 5 mM DTT. Subsequently, free sulfhydryls were protected with iodoacetamide (Sigma-Aldrich, 40 mM, 3 h at 22 °C). Primary amines were then protected by dimethylation using 30 mM formaldehyde (Sigma-Aldrich) and 15 mM NaCNBH₃ (cyanoborohydride, Sterogene) for 16 h at 22 °C. The reductive amination was repeated once for 3 h at 22 °C to ensure full amine protection.

Peptide library purification. Preparative size exclusion chromatography was used first for peptide library purification. The 700-Da exclusion limit of Sephadex G-10 resin (GE Healthcare) allows for desalting as well as the removal of peptides of less than six residues, which would yield cleavage products that do not yield high confidence peptide identifications by MS/MS. Preparations were buffer exchanged to 10 mM potassium phosphate buffer, pH 2.7, 20% (vol/vol) acetonitrile followed by cation exchange chromatography (sulfopropyl Sepharose Fast Flow, GE Healthcare; binding and washing in 10 mM potassium phosphate buffer, pH 2.7, 20% (vol/vol) acetonitrile, elution by 1 M NaCl in 10 mM potassium phosphate buffer, pH 2.7, 20% (vol/vol) acetonitrile), and reversed-phase chromatography (polystyrene-divinylbenzene resin, Polymer Laboratories; washing in 0.3% (vol/vol) formic acid; elution in 80% (vol/vol) acetonitrile). Eluates were concentrated to a minute amount of solvent in a vacuum concentration system (Thermo). Peptide libraries were resuspended in water and stored at -70 °C at concentrations of 2–15 mg/ml. A typical yield was 3 mg of peptide library from 400 ml HEK cell suspension culture at 1.5 × 10⁶ cells/ml.

Cleavage of PICS libraries by test proteases. GluC and bovine thrombin were purchased from Sigma-Aldrich. Neutrophil elastase was purchased from Elastin

Products, cathepsin G was from Calbiochem. HIV protease 1 was a kind gift from Novartis Pharma AG, cathepsin K was a kind gift from D. Bromme (University of British Columbia (UBC), Vancouver, Canada) and caspase 3 and 7 were a kind gift from B. Turk (J. Stefan Institute, Ljubljana, Slovenia). Recombinant human MMP-2 (ref. 44) was activated using 0.1 mM phenyl mercuric chloride⁴⁵. The activated MMP-2 was passed over a gel-filtration spin column (Pierce; the column was pre-equilibrated in 100 mM HEPES 7.4) to remove the organo-mercurial activator and low molecular weight impurities. Buffers for library cleavage were as follows: For GluC, cathepsin G and elastase: 150 mM HEPES, pH 7.4, 50 mM NaCl; for MMP-2 this buffer was supplemented with 5 mM CaCl₂ and 10 μ M ZnCl₂; for HIV protease 1: 100 mM sodium acetate, pH 4.5, 1 mM EDTA, 1 mM DTT; for cathepsin K: 100 mM sodium acetate, pH 5.5, 2.5 mM DTT, 2.5 mM EDTA; for caspase 3 and 7: 50 mM HEPES, pH 7.4, 100 mM NaCl, 1 mM EDTA, 10 mM DTT, 10% (wt/vol) sucrose. Enzyme/library ratios were 1:100 (wt/wt) for thrombin, cathepsin G, neutrophil elastase, HIV protease 1, cathepsin K, caspase 3 and caspase 7. GluC was used at either 1:100 (wt/wt) or 1:1,000 (wt/wt) and MMP-2 was used at a ratio of 1:50 (wt/wt). For thrombin, cathepsin G, neutrophil elastase, MMP-2 and GluC 200 μ g library were used. For HIV protease 1, cathepsin K and caspase 3 and 7, 400 μ g library were used. Assays were performed for 16 h at 37 °C and stopped by heat de-activation (70 °C, 30 min). For HIV protease 1, cathepsin K, and caspase 3 and 7, reactions were buffer-exchanged using reversed-phase solid phase extraction (Sep-Pak, Waters). Eluates were vacuum-evaporated to near dryness and re-solubilized in 200 μ l of 200 mM HEPES, pH 7.4.

Isolation of prime-side cleavage fragments. Neo-amino termini were biotinylated using 0.5 mM sulfosuccinimidyl 2-(biotinamido)-ethyl-1,3-dithiopropionate (amine-reactive biotin with a cleavable disulfide linker, Pierce) and incubated for 2 h at ambient temperature. Biotinylated prime-side cleavage products were separated from the intact peptidic substrates by incubation with Streptavidin Sepharose (GE Healthcare; applied total binding capacity for >150 nmol biotin), 30 min at ambient temperature. After extensive washing, biotinylated cleavage products were eluted in 20 mM DTT (2 h, 22 °C), desalted using reversed-phase solid phase extraction (Sep-Pak, Waters or OMIX tips, Varian,) with binding and washing in 0.3% (vol/vol) formic acid and elution in 80% (vol/vol) acetonitrile. The eluates were vacuum-evaporated to near dryness and stored at -70 °C with a minimum overlay of 2 mM DTT.

LC-MS/MS of isolated prime-side cleavage products. LC-MS/MS analysis was performed on an LC Packings capillary LC system (Dionex) coupled to a quadrupole time-of-flight mass spectrometer (QSTAR Pulsar, Applied Biosystems, operated by the UBC Michael Smith Laboratory/Laboratory for Molecular Biophysics Proteomics Core Facility; or QSTAR XL, Applied Biosystems, operated by the UBC Centre for Blood Research Mass Spectrometry Suite). For the QSTAR Pulsar instrument, samples were resuspended in 2% (vol/vol) acetonitrile, 0.1% formic acid and loaded onto a column packed with PepMap C18 resin (Dionex). Peptides were eluted using a 5–40% gradient of organic phase (buffer B) over 90 min. Buffer A was 2% acetonitrile and 0.1% formic acid, whereas buffer B was 85% acetonitrile and 0.1% formic acid. For the QSTAR XL instrument, samples were resuspended in 5% (vol/vol) acetonitrile, 3% formic acid and loaded onto a column packed with Magic C18 resin (Michrom Bioresources). Peptides were eluted using a 7–40% gradient of organic phase over 95 min. Buffer A was 2% acetonitrile with 0.1% formic acid and buffer B was 98% acetonitrile, 0.1% formic acid. For both instruments, MS data were acquired automatically using Analyst QS software (Applied Biosystems) for information-dependent acquisition based on a 1 s MS survey scan (mass ranges listed below) followed by up to 3 (QSTAR Pulsar) or 2 (QSTAR XL) MS/MS scans of 3 s each. To increase the number of peptide ion spectra collected each sample was typically measured twice to cover complementary mass ranges: 300–700 m/z and 700–1,500 m/z . For several fractions of a cation-exchange fractionated tryptic peptide library (see below) as well as for a small number of test proteases we applied one mass range of 300–1,400 m/z . Nitrogen was used as the collision gas and the ionization tip voltage was 22,000 V (QSTAR Pulsar) or 25,000 V (QSTAR XL).

Multidimensional LC-MS/MS of an intact proteome-derived peptide library.

For cation exchange prefractionation of an intact tryptic peptide library from the soluble lysis fraction of HEK293 cells, ~400 μ g of this library were diluted twofold in 10 mM KH₂PO₄, pH 2.7, 25% (vol/vol) acetonitrile and loaded on a Polysulfoethyl A (Poly LC) 100 $\times$ 4.6 mm, 5 μ m, 300 Å strong cation-exchange column (preequilibrated in 10 mM KH₂PO₄, pH 2.7, 25% (vol/vol) acetonitrile) at 0.5 ml/min, using an AKTA Explorer LC system (GE Healthcare). After washing with 7 ml buffer A, a 20 min gradient was applied to 25% buffer B followed by a 10 min gradient to 100% buffer B. Buffer A was 10 mM KH₂PO₄, pH 2.7, 25% (vol/vol) acetonitrile, 1 M KCl. Fractions were collected every 0.5 ml. Nine fractions were desalted using reversed-phase solid phase extraction. Binding and washing was performed with 0.3% (vol/vol) formic acid and peptides were eluted in 80% (vol/vol) acetonitrile. Samples were vacuum-evaporated to near dryness, stored at 70 °C and analyzed by LC-MS/MS as described above.

MS/MS data analysis. Peak lists of the original QSTAR data were created using Analyst QS software (Applied Biosystems) with the mascot.dll extension (Matrix Science). Peptides were identified from the human IPI database²² by MASCOT³⁴ in conjunction with PeptideProphet⁴⁶ at a 95% confidence level. Search parameters excluded any variable modifications. In accordance with established protocols for QSTAR data analysis^{47,48}, mass tolerance was 0.4 Da for the parent ion and 0.2 Da for fragment ions, allowing up to two missed cleavages. GluC was defined to cleave C terminal to either (i) glutamate or (ii) glutamate and aspartate. Results from both searches were combined. Chymotrypsin was defined to cleave C terminal to tryptophan, tyrosine, leucine and phenylalanine. For prime-side cleavage fragments, semi-style cleavage searches were applied with no constraints for the orientation of the specific terminus and the following static modifications: carboxyamidomethylation of cysteine residues (+57.02 Da), dimethylation of lysine ϵ -amines (+28.03 Da) and thioacylation of amino termini (+88.00 Da). Data originating from the intact tryptic peptide library were searched for fully tryptic peptides with carbamidomethylation of cysteine residues and dimethylation of lysine ϵ -amines and N termini.

Bioinformatic data analysis. Nonprime-side sequences of the original peptidic substrates were inferred from the human IPI database using an in-house Perl software script (**Supplementary Scripts and Note**). The IPI protein sequences are screened for all occurrences of the prime-side cleavage product. At each one, its sequence is extended in the N-terminal direction to include the ten preceding residues. These potential nonprime-side sequences are screened for cleavage sites that match the specificity of the endoprotease used in the library generation. Preceding sequences are only considered for further analysis up to the first occurrence (orientation from C terminus to N terminus) of such a cleavage site.

If the same prime-side fragment originates from different proteins, multiple preceding sequences with potential differences are obtained. In this case, a minimum consensus is built and positions with different residues in different preceding sequences are excluded from further analysis to avoid ambiguity. Excluded from analysis were a small number of peptides (generally fewer than ten) that had N termini corresponding to mature protein termini or were internal peptides with the N-terminal specificity of the endoprotease used for library generation (trypsin, GluC or chymotrypsin). Although these might have been generated by the test protease, both types of peptides can also result from incomplete amine protection during library preparation and were excluded as false positives.

Cleavage site sequence logos. To derive MMP-2 subsite specificities, cleavage site sequences from P4 to P4' (data from peptide libraries generated using the three different endoproteases were kept separate) were treated as multiple sequence alignments (aligned according to subsites P4–P4') and analyzed as sequence logos²¹, applying a published method²⁰ and corrected for the amino acid distribution of the human IPI database. Sequence logos are a powerful tool to discern preferred residues and subsite specificity; especially in cases of broad specificity where canonical consensus motifs inadequately describe selectivity²¹. In sequence logos, the height of the single letter code for amino acids reflects its occurrence rate at each position.

Data and URLs. The peptidic cleavage site sequences accompany this publication as **Supplementary Tables 1–23** and are available in MEROPS⁵ (<http://merops.sanger.ac.uk>), the peptidase database. The original mass spectrometer files are available online at PRIDE⁴⁹ (<http://www.ebi.ac.uk/pride/>) with accession numbers listed in **Supplementary Table 24**. Additional software tools for analysis of PICS cleavage site sequences, including heat map generation, are available at <http://www.clip.ubc.ca/resources/index.html>. A detailed experimental protocol can also be found at http://www.natureprotocols.com/2008/05/14/protease_subsite_profiling_wit.php.

Note: Supplementary information is available on the Nature Biotechnology website.

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AUTHOR CONTRIBUTIONS

O.S. designed and implemented the chemistry in the workflow, performed all analyses, wrote the Perl scripts and performed the bioinformatics and wrote and edited the paper. C.M.O. conceived, designed and oversaw the development of PICS, performed proof of concept experiments, wrote and edited the paper and provided financial support for the project.

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